Predator-specific deployment of defensive behaviors in the marine isopod *Cleantiella isopus* (Isopoda, Idoteidae): Arched posture against fish and escape against crabs

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Running header: Defensive behaviors in a marine isopod

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Abstract

Many prey species are exposed to multiple predators that use contrasting hunting modes. To enhance survival under such conditions, the prey may discriminate among predator types and deploy predator-specific defenses. Secondary defenses expressed following contact with predators are likely to require an accurate assessment of predation risk, as inappropriate responses can prove fatal. However, the selective deployment of secondary defenses against multiple predators remains poorly understood. Tonic immobility (thanatosis or death-feigning) is one of the most widely reported secondary defenses of diverse taxa. The marine isopod *Cleantiella isopus* exhibits tonic immobility in response to predator attacks, adopting a rigid, arched posture while remaining motionless. Here, we tested whether tonic immobility was selectively deployed against different predator types by comparing isopod responses to three fish species and a crab. Tonic immobility was exclusively expressed in response to fish predators, whereas crabs elicited escape responses. Further experiments revealed that the arched posture reduced the probability of swallowing, whereas immobile individuals were attacked less frequently and experienced longer delays before the attack. These results suggest that tonic immobility in *C. isopus* enhances survival against fish predators through multiple mechanisms. Our findings demonstrate that tonic immobility in *C. isopus* is not a generalized response to external stimulation, but rather a predator-specific defense against fish predators, supporting the hypothesis that tonic immobility represents an adaptation to particular predator types. The ability to discriminate between predator types and selectively deploy appropriate defensive behaviors may enable *C. isopus* to enhance its survival in environments with multiple predators.

Keywords

antipredator behavior, crustacean, intertidal, predation, predator avoidance

Introduction

Predation is a major selective pressure that shapes the evolution of diverse life-history, morphological, and behavioral traits that reduce both predator encounters and predation success (Endler, 1986; Ruxton et al., 2018). These defenses are thought to function at different stages of the predation sequence (i.e., detection, identification, approach, subjugation, and consumption; Endler, 1991). Prey species typically rely on primary defenses that minimize encounters with or detection by predators, often through the use of chemical cues in aquatic systems (Ferrari et al., 2010; Kats and Dill, 1998). In contrast, secondary defenses enhance survival during the subjugation and consumption stages of the predation sequence (Ruxton et al., 2018). For example, some prey can escape even after being taken into predators' mouths, as reported for hairworms and juvenile eels that emerge from the gills of fish (Ponton et al., 2006; Hasegawa et al., 2022), whereas carabid beetles can chemically counterattack after being swallowed by frogs (Sugiura and Sato, 2018).

Many prey species are exposed to multiple predators that use contrasting hunting modes (Sih et al., 1998). To persist in environments where multiple predator types coexist, prey are expected to evolve multiple defenses tailored to predators' hunting modes (Kikuchi et al., 2023; Kishida and Nishimura, 2005; Rojas et al., 2017). Because secondary defenses are often deployed as a "last resort" (Ruxton et al., 2018), inappropriate expression against the wrong predator type may be fatal. Consequently, the ability of the prey to discriminate between approaching predators and deploy predator-specific secondary defenses can be under strong selection. However, most studies demonstrating predator-specific defenses against multiple predators have examined prey responses to the presence of predators or their cues before direct contact (e.g., Relyea, 2001; Alcaraz and Arce, 2017), or have shown that different defensive responses are elicited because different predators attack different body parts, rather than reflecting the active discrimination of predator types by the prey (Rojas et

al., 2017). Therefore, empirical evidence remains limited on whether prey can discriminate among predators at close range, just prior to consumption, and deploy the most effective defense against each predator type.

Tonic immobility (also termed thanatosis or death feigning) is a widely reported behavioral secondary defense across various taxa, in which prey remain motionless and adopt characteristic postures in response to predatory stimuli (Humphreys and Ruxton, 2018; Miyatake, 2026; Sakai, 2021). This is thought to involve multiple potentially interacting mechanisms through which tonic immobility enhances prey survival, rather than solely through the mimicry of death (Miyatake et al., 2009; Ruxton, 2006). For example, visual predators such as spiders are less likely to attack immobile prey; therefore, immobility can increase survival in encounters with visually oriented predators (Miyatake et al., 2009; 2004). In some systems, the defensive posture itself, such as extending the body spines, may confer protection as a form of physical defense, even in the absence of prolonged immobility (Honma et al., 2006; Moore and Williams, 1990). However, experimental evidence for the predator-avoidance mechanisms of tonic immobility remains largely restricted to a limited subset of terrestrial insects (but see Yamada and Urabe, 2021).

Although studies on marine systems are limited compared to the extensive body of work on terrestrial systems, tonic immobility has been repeatedly reported in marine environments, particularly among crustaceans (e.g., Bergey and Weis, 2006; Field, 1990; Harada et al., 2025; Mackay, 1929). A previous study reported that the marine isopod *Cleantiella isopus* (Isopoda, Idoteidae) exhibits tonic immobility in response to attacks by fish (Figure 1; Igarashi and Wada, 2024). This is a common intertidal benthic species inhabiting intertidal and shallow subtidal rocky shores in East Asia, and is typically found under rocks (Takahashi and Goshima, 2012). During tonic immobility, it assumes an arched posture, raising its head and tail, and remains rigid for 1–5 min (Figure 1). *C. isopus* exhibits five

distinct color morphs (Takahashi and Goshima, 2012), and lateral body coloration becomes exposed during tonic immobility, whereas individuals remain immobile on algal or rocky substrates. The frequency and duration of tonic immobility do not differ among color morphs, although smaller individuals exhibit longer durations of tonic immobility (Igarashi and Wada, 2024). *C. isopus* also exhibits tonic immobility in response to artificial stimuli, such as poking with sticks (Igarashi and Wada, 2024). If this behavior is expressed indiscriminately, it could suggest either that tonic immobility is effective regardless of the predator hunting mode or that this species has a limited ability to discriminate among predators.

Marine isopods are preyed upon by fish and crustaceans in contrasting hunting modes (Wallerstein and Brusca, 1982; Leidenberger et al., 2012; Yli-Renko et al., 2022), and species in the genus *Cleantiella* have been recorded as prey for both coastal fish and crustaceans (Shin and Ryu, 2025; Shiraki and Kakui, 2022). In particular, *C. isopus* coexists with fish, which rely on vision to detect and capture prey by swallowing, and crabs, which rely on olfaction to capture prey with their claws. These two predator types can differ markedly in the selective pressures imposed on prey behavior.

In this study, we investigated tonic immobility in the marine isopod *C. isopus*. We hypothesized that tonic immobility is preferentially expressed in fish predators and contributes to their survival. In Experiment 1, we tested whether defensive responses differed among predator types with contrasting hunting modes. We exposed *C. isopus* to two types of predators: three intertidal fish species and the varunid crab *Gaetice depressus* (Decapoda, Varunidae). We predicted that tonic immobility would be expressed consistently across multiple fish species but not in response to crabs. In Experiment 2, we examined two possible functional explanations for the antipredator effects of tonic immobility. First, we predicted that predators would take longer to attack immobile individuals because fish predators are generalist predators that primarily feed on live prey and may therefore be less likely to attack

immobile prey. Second, if the arched posture provided a physical defense, we predicted that predators would be more likely to reject isopods displaying this posture.

Materials and Methods

Prey and predator collection and maintenance

Juvenile isopods were used in all experiments because their small body size makes them particularly vulnerable to predation by fish and other potential predators. Therefore, tonic immobility may be especially important for juveniles, and smaller individuals of *Cleantiella isopus* have been shown to exhibit longer durations of tonic immobility (Igarashi and Wada, 2024). Similar to other isopods, *C. isopus* females lay eggs in a brood pouch on their abdomens. We collected ovigerous females from an intertidal rocky shore in Kattoshi (41°44′N, 140°36′E), southern Hokkaido, Japan, during low tides in May and July 2022, the breeding season of *C. isopus* (Takahashi and Goshima, 2012). Ovigerous females were maintained in 21 L plastic containers (water depth: 5 cm) under a 12:12-h light:dark photoperiod (daytime: 0800–2000; nighttime: 2000–0800) at ambient temperature (11–13 ℃). The containers were maintained with field-collected seawater, and both the seawater and food (dried kelp, *Undaria pinnatifida*) were replaced once daily. Juveniles hatched from ovigerous females were reared individually under the same conditions until further use.

The study site was a broad rocky platform inhabited by various benthic predators that feed on small invertebrates. In the first experiment, we used four predator species representing two feeding types: predators that swallowed prey whole and predators that gnawed on prey. The former comprises three species of fish: the forktongue goby *Chaenogobius annularis* (Perciformes, Gobiidae), the eelpout *Zoarchias veneficus* (Perciformes, Stichaeidae), and the elegant sculpin *Bero elegans* (Perciformes, Psychrolutidae). We used multiple fish species because fish can differ substantially in morphology and feeding strategies, which may

influence the effectiveness of prey behavior. *C. annularis* is one of the most common intertidal fish species in Japan and is widely distributed throughout East Asia (Murase et al., 2017). *Z. veneficus* and *B. elegans* are distributed throughout the northwestern Pacific, including northern Japan (Masuda et al., 1984). Although the composition of potential fish predators may vary geographically, gobies, eelpouts, and sculpins are widely distributed intertidal fishes and are likely to represent common predator groups encountered by *C. isopus* across much of its range. These species coexist with *C. isopus* at our study site but differ in their microhabitats: goby populations are the densest and most widespread, sculpins occur only on reefs, and eelpouts occur only on seaweed (KI personal observation). The forktongue goby actively searches for prey (Nakayama et al., 2025), whereas the eelpout has a relatively small mouth and mainly feeds on mobile prey on seaweed (Sakurai et al., 2009). In contrast, the elegant sculpin has the largest mouth relative to body size among the three species and is a typical sit-and-wait predator, as is common in Cottoidea sculpins (e.g., Laurel and Brown, 2006; Horinouchi et al., 2009). All three fish species belonged to groups that are generally recognized as generalist predators of benthic invertebrates in intertidal habitats (Sasaki and Hattori, 1969; Yoshiyama, 1980; Sakurai et al., 2009).

The latter type was represented by the varunid crab *Gaetice depressus*, which is often found under boulders, along with *C. isopus*. This species is one of the most common intertidal crabs in the northwestern Pacific, particularly in East Asia (Davie and Ng, 2007). Although *G. depressus* is generally regarded as an omnivorous crab (Wahyudi et al., 2013), it has also been recognized as an active predator of live benthic crustaceans (Yoshino et al., 2002; Hamasaki et al., 2023). Other potential predators, such as nemerteans and shorebirds, may also occur in coastal ecosystems. However, this study focused on fish and crabs because they were far more common at our study site.

In the second experiment, we used the forktongue goby, which is the most abundant

predatory fish at our study site. All predators were collected from the same sites as the isopods using hand nets. They were maintained under a 12:12-h light:dark cycle and fed sinking pellets (Hikari Crest Cat, radius < 5 mm; Kyorin Co., Ltd., Himeji, Japan) for several days before the experiment. To ensure sufficient motivation for foraging, the predators were not fed for 2 days before the experiments.

Before the experiments, we measured the body sizes of the prey and predators individually from dorsal photographs using Fiji software (Schindelin et al., 2012). Isopod body size was measured as the distance from the tip of the head to the end of the tail (Takahashi and Goshima, 2012), fish size as the total length from the tip of the snout to the end of the caudal fin, and crab size as carapace width. In all experiments, each predator–prey pair was used only once and was not reused in subsequent trials. This study did not require ethical approval from human or animal welfare committees. All the predators and surviving prey returned to their original locations after the experiment.

Experiment 1: Responses to different predator types

We conducted predation trials to determine whether tonic immobility defends against fish predators and whether isopods can distinguish between different predator types. Juvenile *C. isopus* used in this experiment (n = 89; body length: 2.9–5.7 mm). The three fish species represented predators that swallow prey whole: the forktongue goby *C. annularis* (n = 19; total length [TL]: 41.5–61.5 mm), the eelpout *Z. veneficus* (n = 27; TL: 52.6–81.3 mm), and the sculpin *B. elegans* (n = 24; TL: 15.7–44.4 mm). The varunid crab *G. depressus* (n = 19; carapace width: 4.8–11.2 mm) represented predators that gnaw on prey. Fish predators were housed individually in opaque plastic containers (8.8 × 18 × 7 cm, seawater depth: 5 cm) with a black-bottomed substrate for 1 h before the trials. Trials with crabs were conducted in opaque plastic containers (9.5 cm × 13 cm × 6.5 cm, seawater depth: 4 cm) with a black-

bottomed substrate. After 1 h of acclimatization, we gently released a juvenile isopod into the end of the container opposite the predator. We recorded the interactions for 10 min (crabs for 30 min) using a camera (JVC, EverioR GZ-F720, positioned 50 cm directly above the container). Preliminary observations indicated that fish predators typically contacted or attacked the prey within 10 min of prey introduction. In contrast, crabs interacted with prey more slowly, and interactions often did not occur within 10 min; therefore, crab trials were observed for 30 min. Videos were analyzed using the JWatcher software (Blumstein and Daniel, 2007) to quantify tonic immobility, fleeing, and predation. Fish trials were conducted in July 2022, and crab trials in April 2023. For the fish trials, we additionally recorded whether the predators attacked the isopods after the onset of tonic immobility.

Experiment 2: Demonstrating the function of tonic immobility

We used the forktongue goby *C. annularis* (TL: 40.1–75.0 mm) as a predator and juvenile *C. isopus* (body length: 3.1–7.1 mm) as prey. Predator–prey pairs were assigned to either a control treatment (n = 59) or a freshwater treatment (n = 60). In the freshwater treatment, passive individuals were generated by placing juvenile isopods in freshwater for 1 h prior to the experiment, which induced reduced activity or immobility for several hours owing to osmotic stress (Jormalainen and Merilaita, 1995; Miura and Goshima, 2016). We tested the possible antipredator functions of tonic immobility by comparing the latency to attack and survival between passive individuals that remained immobile throughout the trial and control individuals that remained active until attack and then exhibited tonic immobility. We predicted that (1) the latency until attack would be longer for passive individuals than for control individuals and (2) control individuals would have a higher survival rate because they could adopt tonic immobility after an attack.

Gobies were housed individually in opaque containers (8.8 cm × 17.9 cm × 7.0 cm,

seawater depth: 5 cm) with a black-bottomed substrate for 1 h before the trials. At the start of each trial, we released a juvenile isopod at the end of the container opposite the predator and recorded the interactions for 10 min using a camera (JVC, EverioR GZ-F720, positioned 50 cm directly above the container). Preliminary observations indicated that gobies typically attacked prey within 10 min of introduction. From the video recordings, we measured the latency to the first attack (s) and the predation frequency. Each predator and prey was used only once. The experiments were conducted in July 2022.

Statistical analyses

All statistical analyses were performed using R v. 4.5.1 software (R Core Team, 2025). In Experiment 1, pairwise Fisher's exact tests with Holm's method were conducted using the "fisher.multcomp" function in the "RVAideMemoire" v. 0.9.83.12 package (Herve, 2025) to assess whether the frequencies of defensive behaviors differed among predator species, under the null hypothesis of no differences in defensive behavior across multiple predators. Two contrasting defensive behaviors—tonic immobility and fleeing—were observed, and comparisons focused on differences in their frequencies across the predator species. To examine whether the predation frequency differed among predator species, we conducted a pairwise Fisher's exact test with Holm's correction.

To further examine whether the immobile state during tonic immobility reduced the likelihood of subsequent attacks by fish predators, we restricted the analyses to trials in which individuals exhibited tonic immobility. The response variable was whether individuals were attacked (attacked = 1, not attacked = 0), and the explanatory variable was the behavioral state of the isopod (active vs. tonic immobility). To account for potential differences among predator species, we included predator species and predator size as fixed effects and incorporated trial ID as a random effect to control for the non-independence of observations

within trials. We fitted a binomial generalized linear mixed-effects model (GLMM) with a logit link using the "glmmTMB" v. 1.1.11 package (Brooks et al., 2017).

In Experiment 2, our primary goal was to test whether prey adopting tonic immobility had a survival advantage over those that did not. We constructed a binomial generalized linear model (GLM) with survival as the response variable, treatment as the explanatory variable, and predator body size as the covariate. This covariate was included to control for variation in predator size among the trials, as predation success in gape-limited predators depends on relative body size. We assessed the statistical significance of the explanatory variables using likelihood ratio tests via the "Anova" function in the "car" v. 3.1.3 package (Fox and Weisberg, 2019). For plotting results, we calculated estimated marginal means and 95% confidence intervals using the "emmeans" function in the "emmeans" v. 1.11.2 package (Lenth, 2025).

To further investigate whether maintaining an immobile state reduces the likelihood of predator attacks, we fitted a Cox proportional hazards model using the "coxph" function from the "survival" v. 3.8.3 package (Therneau, 2024). We treated trials without attacks during the 10-min observation period as censored. We included the predator body size as a covariate in the model. The proportional hazards assumption was satisfied, as assessed with the "cox.zph" function ($P > 0.05$). Adjusted survival curves, accounting for covariates, were generated to visualize the time-dependent detection risk for each treatment.

Results

Experiment 1: Responses to different predator types

The defensive behaviors exhibited by the isopods varied significantly among the predator species (Figure 2). Isopods displayed tonic immobility exclusively in response to fish predators, whereas they struggled and attempted to flee when encountering crabs. Pairwise

comparisons indicated that the frequency of defensive behaviors against crabs differed significantly from those against all fish predators (Fisher's exact test, Holm-adjusted *P*-values; vs. goby: $P < 0.001$; vs. eelpout: $P < 0.001$; vs. sculpin: $P = 0.002$), but there were no significant differences among the fish predators themselves. Isopods also exhibited tonic immobility in the absence of attacks, induced solely by incidental physical contact with the predator during movement or by water movement, particularly in tests with eelpouts (51.9%) and sculpins (20.8%) (Figure 2). The predation frequency was significantly lower in eelpouts than in other predator species (Fisher's exact test, Holm-adjusted *P*-values; vs. goby: $P = 0.044$; vs. sculpin: $P = 0.001$; vs. crab: $P = 0.022$).

Exhibiting tonic immobility significantly reduced attack probability, and this effect differed among predator species (Table 1). The model predictions indicated that attacks on immobile individuals were particularly infrequent in eelpouts and sculpins, whereas gobies frequently attacked them (Figure 3).

Experiment 2: Demonstrating the function of tonic immobility

The second experiment used 119 trials (control: $n = 59$, freshwater [induced passivity]: $n = 60$); 115 involved predator–prey interactions (control: $n = 58$, freshwater: $n = 57$). Individuals in the control treatment exhibited tonic immobility when attacked by gobies and were fully engulfed in their mouths, thereby avoiding predation and being spat out, resulting in survival odds approximately 2.7 times higher (Figure 4a; Table 2).

However, individuals in the freshwater treatment that remained immobile took longer to be attacked by gobies than mobile individuals in the control treatment. The hazard ratio for the freshwater treatment relative to the control was 0.63 (Figure 4b; Table 3), indicating that the isopods in the freshwater treatment were less likely to be attacked by gobies at any given time.

Discussion

This study demonstrates that tonic immobility in the marine isopod *C. isopus* is a predator-specific defensive response against fish rather than a generalized response to tactile stimulation. In Experiment 1, isopods exhibited tonic immobility only when attacked by fish, whereas attacks by crabs elicited struggling and escape behaviors. In Experiment 2, individuals that adopted the arched posture had a higher probability of surviving fish attacks. In contrast, individuals that became immobile without adopting the arched posture were less likely to be attacked but were readily consumed once attacked. Taken together, these results indicate that tonic immobility in *C. isopus* is selectively deployed against fish predators, and that the arched posture and subsequent immobility enhance survival. The arched posture characteristic of *C. isopus* may be more widespread among idoteid isopods than is currently recognized. Given that idoteid isopods are commonly preyed upon by fishes (Wallerstein and Brusca, 1982; Leidenberger et al., 2012) and share broadly similar body plans, and that tonic immobility in *C. isopus* consists of a simple response—rigid immobility in an arched posture—this form of tonic immobility is likely to occur in multiple species within Idoteidae.

The ability to discriminate among predator types is widely recognized in aquatic systems (e.g., Alcaraz and Arce, 2017; Kishida and Nishimura, 2005; Lakowitz et al., 2008), and our study provides further evidence that prey can discriminate between approaching predator types during the later stages of the predation sequence (i.e., subjugation and consumption) and deploy distinct defensive behaviors accordingly. Switching between alternative defensive behaviors against each predator type may allow *C. isopus* to enhance its survival in environments where multiple predators coexist. In aquatic systems, predator recognition has traditionally been assumed to rely primarily on the waterborne chemical cues released by predators (Ferrari et al., 2010). However, a previous study has suggested that

proximal cues, such as direct physical contact, may play an important role in assessing predation risk, even in aquatic environments (Okamiya and Kishida, 2022). Because tonic immobility in *C. isopus* can be induced solely by tactile stimulation in the absence of chemical cues (Igarashi and Wada, 2024), tactile information likely plays a key role in predator discrimination in this species. In particular, crab predators typically impose sustained tactile stimulation while restraining the prey, whereas fish attacks involve brief, instantaneous contact. Isopods may exploit this contrast to discriminate between the predator types. However, tonic immobility was occasionally induced in response to incidental physical contact or water flow generated by fish, even in the absence of an attack. Moreover, although the crabs produced comparable levels of physical contact, such stimuli rarely elicited tonic immobility. These observations suggest that predator discrimination in *C. isopus* is not determined solely by specific attack patterns. Prey often integrate multiple sensory modalities when assessing predation risk rather than relying on a single cue (Ode and Wissinger, 1993; Yamamoto et al., 2023). Because the defensive responses observed here occurred after close contact with predators, multiple cues were likely available simultaneously. Thus, *C. isopus* may rely on multimodal predator discrimination by combining tactile information with other cues such as olfactory or visual signals. Identifying the sensory mechanisms underlying predator discrimination in this species is an important avenue for future research.

Several hypotheses have been proposed to explain how tonic immobility enhances prey survival (Miyatake et al., 2009). These include the “concealment” or “background matching” hypothesis, in which immobility reduces detectability (Ohno and Miyatake, 2007); the “physical defense” hypothesis, in which postural changes increase effective body size and hinder handling (Honma et al., 2006); and the “loss of predator interest” hypothesis, in which predators stimulated by movement lose interest in immobile prey (Miyatake et al., 2004). Our findings suggest that tonic immobility in *C. isopus* operates through multiple interacting

326 mechanisms. Individuals exhibiting tonic immobility were more likely to survive predator
327 attacks, suggesting that adopting an arched posture reduces the likelihood of ingestion by
328 gape-limited fish predators. In gape-limited predators, the size of prey that can be consumed
329 is constrained by body depth (Brönmark and Miner, 1992; Nilsson and Brönmark, 2000).
330 Tonic immobility in *C. isopus* increases the effective body depth through a postural shift,
331 which likely makes it more difficult for fish to handle and ingest. This interpretation is
332 consistent with previous studies demonstrating the predator-avoidance function of defensive
333 postures during tonic immobility (Honma et al., 2006; Moore and Williams, 1990), although
334 *C. isopus* is unusual in that it lacks conspicuous morphological defenses, such as spines. This
335 posture may be particularly important for smaller individuals, such as those used in the
336 present study, because smaller prey are often more vulnerable to fish predators (e.g., Stein and
337 Magnuson, 1976; Nilsson and Brönmark, 2000). Consistent with this possibility, our previous
338 study showed that smaller individuals exhibited longer durations of tonic immobility (Igarashi
339 and Wada, 2024), suggesting a greater investment in this behavior.
340 Our results also show that immobile individuals were less likely to be attacked and
341 experienced longer delays before being attacked by fish predators. There are several possible
342 explanations for these results. First, sustained immobility may reduce the detectability of fish
343 predators (Ohno and Miyatake, 2007). Marine isopods of the family Idoteidae often exhibit
344 cryptic coloration through disruptive coloration and background matching (Hultgren and
345 Mittelstaedt, 2015; Merilaita, 1998), whereas *C. isopus* shows similar coloration patterns
346 (Takahashi and Goshima, 2012). Under natural conditions, adopting an arched posture may
347 expose the lateral body coloration and enhance background matching with algal or rocky
348 substrates, thereby reducing the likelihood of an attack. In addition, an arched posture may
349 cause predators to mistake individuals for non-prey items, such as drifting algal debris. As our
350 experiments were conducted against a uniform black background, rather than within

heterogeneous habitats occupied by *C. isopus*, predator attacks may have been more frequent than under natural conditions. However, these possibilities were not directly evaluated in the present study. Second, sustained immobility may fail to elicit predatory motivation in predators that preferentially respond to moving prey (Miyatake et al., 2004, 2009). This explanation is consistent with the tendency of fish to respond more strongly to moving prey than to stationary prey (Rowland, 1999), as well as with our observation that fish predators rarely respond to immobile isopods unless they resume movement. Third, immobile individuals may have been perceived as low-profit prey (Ruxton, 2006). Predators may avoid prey that require additional handling time because such prey are less profitable from an optimal foraging perspective (Schoener, 1969; Stephens et al., 2008). An arched posture may increase handling costs, thereby reducing prey profitability. Our results indicate that tonic immobility contributes to survival against predation in *C. isopus*; however, the underlying mechanisms warrant further investigations.

The effectiveness of tonic immobility in *C. isopus* also appears to differ among fish predators. The predation frequency was significantly lower in eelpouts, whereas gobies and sculpins exhibited comparatively high predation success rates. In particular, most individuals attacked by sculpins were consumed. In contrast, many individuals that survived encounters with sculpins had already entered tonic immobility prior to the attack, which was often triggered by incidental physical contact or water movement generated during the predator's approach. These results suggest that the defensive effect of the arched posture may be limited to fish predators with relatively large mouths. This interpretation is also consistent with the relatively small mouth size of eelpouts compared with that of gobies and sculpins, because the increase in handling costs caused by the arched posture may have a stronger effect on predators with smaller gape sizes. In contrast, gobies frequently re-attacked immobile individuals, which may reflect the broad feeding habits of this species (Sasaki and Hattori,

1969). Together, these results indicate that tonic immobility in *C. isopu*s is not uniformly effective against all fish predators but instead confers context-dependent benefits depending on predator morphology and behavior.

This study demonstrated that tonic immobility, characterized by an arched posture, has an antipredator function in the marine isopod *C. isopus* and is selectively expressed depending on the predator type. The ability to switch among alternative defensive behaviors according to predator type has long been recognized across diverse taxa (e.g., Aizaki and Yusa, 2010; Alcaraz and Arce, 2017; Bourdeau, 2009), but our results show that such discrimination and switching defensive behaviors can occur even at late stages of the predation sequence, when prey are already under attack and close to consumption. Our findings also indicate that tonic immobility, despite being a relatively simple behavioral response, can contribute to prey survival through multiple interacting mechanisms. Future studies should clarify the sensory cues underlying predator recognition, including whether tactile cues alone are sufficient or whether multiple sensory modalities are integrated. In addition, determining the prevalence of arched-posture tonic immobility within Idoteidae is important for understanding how predator-specific selection pressures have shaped the evolution of distinctive antipredator behaviors. In coastal ecosystems, where fish represent the dominant predators, similar forms of tonic immobility may be widespread among crustaceans, as suggested by previous observations (e.g., Field, 1990; Hazlett and McLay, 2005). More broadly, our findings emphasize that evaluating the effectiveness of prey defenses across multiple ecologically relevant predators is essential to fully understand the selective forces shaping antipredator behavior.

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Competing interests

The authors declare no competing interests.

Author contributions

Koichi Igarashi: Conceptualization, Methodology, Formal analysis, Investigation, Writing – original draft, Writing – review & editing, Visualization. **Satoshi Wada**: Writing – review & editing, Resources, Supervision.

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Figure legends and table captions

Figure 1

The marine isopod *Cleantiella isopus*. (A) Dorsal view under normal state and (B) lateral view during tonic immobility (scale bar = 10 mm).

Figure 2

Relative frequencies of defensive behaviors exhibited by *C. isopus* in response to different predator species. TI = tonic immobility; TI with attack = tonic immobility induced by predator attack; TI without attack = tonic immobility induced by incidental contact with the predator; Fleeing = escape response following attack; Predation = prey captured and consumed by the predator.

Figure 3

Estimated probabilities of fish predator attack on isopods in an active state or exhibiting tonic immobility. Columns represent estimated marginal means from the fitted model, and error bars show 95% confidence intervals.

Figure 4

(A) Estimated marginal means of survival probability with 95% confidence intervals (error bars) for each treatment group, obtained from the fitted model. (B) Survival curves from the Cox proportional hazards model showing time to first predator attack for each treatment group.

Table 1

Summary of the GLMM analyzing differences in attack probability between active and tonic immobility states in Experiment 1. Goby and active state were used as reference categories.

601 Significant values ($P < 0.05$) are shown in bold.

602

603 **Table 2**

604 Summary of the binomial GLM analyzing differences in the survival probability between

605 treatments in Experiment 2. Control treatment was used as the reference category. Significant

606 values ($P < 0.05$) are shown in bold.

607

608 **Table 3**

609 Summary of the Cox proportional hazards model analyzing time to first predator attack in

610 Experiment 2. Control treatment was used as the reference category. Significant values ($P <$

611 0.05) are shown in bold.